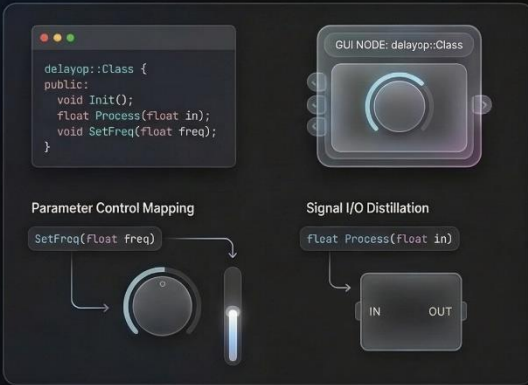


Fluid Spectra: Engineering the Liquid Glass UI

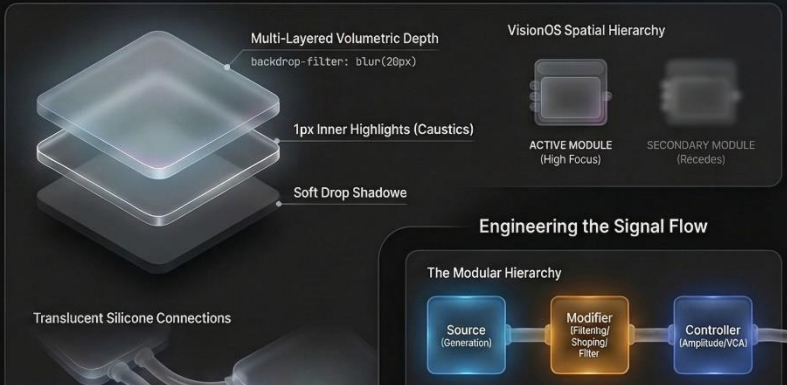
The Visual Grammar (Mapping the C++ API to Spatial Nodes)



Luminous Semantic Color Coding



The "Liquid Glass" Material Stack



Engineering the Signal Flow

The Modular Hierarchy



Active Signal Visualization



NotebookLM

Fluid Spectra: Engineering the Spatial Liquid Glass UI

From Flat Schematics to Spatial Computing. Moving beyond concept-art clichés to a production-viable, ergonomic software design inspired by VisionOS and high-end WebGL dynamics.

BASELINE: FLAT NEON GLASSMORPHISM



UPGRADE: SPATIAL VOLUMETRIC LIQUID GLASS UI



NotebookLM

The Dynamic Inspector Drawer

Translating the C++ backend into a tactile workspace.

Deep parameter editing is handled in a unified, slide-out drawer, keeping the main canvas strictly focused on signal routing.



daisysp::ReverbSc

SetFeedback



85%

SetLpFreq



2.5 kHz

SetDamping



0.70



Noise .M



OVPE

Pod Step Sequencer (Enhanced)

Hardware

Export C++

MODULES

Search modules...

SOURCES

Oscillator

Waveform generator

SEQUENCERS

8-Step Sequencer

CV step sequence

PROCESSORS

Moog Filter

CV filter

MODULATORS

VCA

Voltage controlled amp

EFFECTS

Envelope

ADSR / AD generator

MODULATORS

LFO

Low frequency osc

EFFECTS

Delay

Variable delay line

EFFECTS

Reverb

Sharp room/hall

EFFECTS

Overdrive

Soft clipping

EFFECTS

Wavefolder

Wavefolding shaper

EFFECTS

Bitcrush

Bit depth reduction

T/D & CONTROL

Output

Audio output to DAC

T/D & CONTROL

Knob

Hardware knob input

T/D & CONTROL

Encoder

Rotary encoder input

T/D & CONTROL

LED

Hardware LED output

Knob modules to connect to add

KNOB 1

Decay/Time

Chan: 0

Min: 0.01

OUT

ENCODER

Start/Stop

Pin: 0

PWR: true

OUT

LED 1

Start/Stop

Pin: 0

PWR: true

8-STEP SEQ

Trac: 0

Tempo: 120 BPM

CV

GATE

TRIG

TEMPO CLK

Freq: 3.00 Hz

OUT

ENVELOPE

Attack: 0.01s

Decay: 0.50s

TRIG

DEC CV

OUT

OSCILLATOR

Freq: 110 Hz

Ampl: 1.00

TRIG CV

OUT

VCA

Gain: 1.00

IN

CV

OUT

MOOG FILTER

Cutoff: 20000 Hz

Reso: 0.70

TRIG CV

OUT

OUTPUT

Gain: 0.00

OUT

KNOB 2

Position/Value

Chan: 1

Min: 0.00

OUT

LED 2

Start/Stop

Pin: 1

PWR: true

IN

INSPECTOR

×

MODULE

Label

ENVELOPE

DaisySP Class

daisyv2::ADEnv

Category

Modulators

TIME

Attack

0.01s

Decay

0.50s

SHAPE

Curve

0.00

Envelope curve shape (-10 to +10)

CONNECTIVITY

INPUTS

TRIG

Trigger

→ 8-STEP SEQ

DEC CV

IN

→ KNOB 1

OUTPUTS

OUT

OUT

→ 2 com.

DOCUMENTATION

The ENVELOPE provides standard ADSR generator for amplitude and modulation control. Connect trigger inputs to gate sources for rhythmic envelope shaping.

Audio waveform preview

OVPE

v2.0 neon

Dual-Oscillator Subtractive Synthesizer

Library

Settings

Compile

SOUND SOURCES

OSCILLATOR

OSC 1

Wave

Freq: 440 Hz

Detune: 0 ct

Level: 0.8

V/OCT

FM

OUT

OSCILLATOR

OSC 2

Wave

Freq: 440 Hz

Detune: +7 ct

PW: 50%

V/OCT

PWM

OUT

UTILITY

MIXER

OSC 1: 0.7

OSC 2: 0.5

Noise: 0.1

IN 1

IN 2

OUT

SOUND SHAPING

FILTER

VCF

Type: LP

CUTOFF: 2.4 kHz

RESO: 0.6

ENV: 0.45

IN

CV

OUT

ENVELOPE

ENV FILTER

A: 15ms

D: 120ms

S: 0.3

R: 250ms

GATE

OUT

ENVELOPE

ENV AMP

A: 5ms

D: 80ms

S: 0.8

R: 400ms

GATE

OUT

MODULATION & OUTPUT

MODULATOR

LFO

Wave

Rate: 2.5 Hz

Depth: 0.45

OUT

AMPLIFIER

VCA

Gain: 1.0

IN

CV

OUT

HARDWARE

DAISY OUT

Master: 0.75

SR: 48 kHz

Buf: 256

L

R

OVPEPod Step Sequencer (Enhanced)

HardwareExport S++

INSPECTOR

ENCLOPE

Label

Category

Attack0.00s

Decay0.50s

Curve0.00

Envelope curve slope (-10 to +10)

TRIGTrigger4-STEP SEQ

SEC CVUpNOISE 1

NOTOff+ 2 comp.

DOCUMENTATION

The ENVELOPE provides standard AD/AR/DSR generator for amplitude and modulation control. Connect trigger inputs to gate sources for rhythmic envelope shaping.

Module Overview Preview

MODULES

Search modules...

SOURCES

OSCILLATOR

Tempo Clock

PROCESSORS

B-Step Sequencer

Moog Filter

VCA

MODULATORS

Envelope

LED

EFFECTS

Delay

Reverb

Overdrive

Wavefolder

Bitcracker

T/B & CV/TOOLS

Output

Knob

Encoder

LED

Knob 1

Encoder

LED 1

LED 2

Tempo CLK

B-STEP SEQ

Envelope

Oscillator

VCA

Moog Filter

Output

B-STEP SEQ